

MTH203: Sequences and series

Homework 02

26/08/2025, 6:00 PM

Problem 1. Compute the supremum of the set

$$S := \{n/(3n + 1) : n \in \mathbb{N}\}.$$

Problem 2. Let $A \subset \mathbb{R}$ be nonempty and bounded above, and let $s \in \mathbb{R}$ have the property that for all $n \in \mathbb{N}$, $s + 1/n$ is an upper bound for A and $s - 1/n$ is not an upper bound for A . Show $s = \sup(A)$.

Problem 3. Prove that $\bigcap_{n=1}^{\infty} (0, 1/n) = \emptyset$.

Problem 4. Let A_1 and A_2 be subsets of $\mathbb{R}$ that are bounded above. Find a formula for $\sup(A \cup B)$ in terms of $\sup(A)$ and $\sup(B)$.

Problem 5. Let A be a bounded above subset of $\mathbb{R}$. Show that if $c \geq 0$, then $\sup(cA) = c \sup(A)$.